

Modeling How Macroeconomic Shocks Affect Regional Employment: Analyzing the Brazilian Formal Labor Market using the Global VAR Approach.

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Talking points

- 1 CEMAP:
- 2 Motivation
- 3 Challenges and Methods
- 4 Data
- 5 Our Results
- 6 Limitations and Possible Extensions
- 7 A closer look on the benefits of Automatic Selection
- 8 Conclusions

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Center for Applied Macroeconomic Research - FGV

- Center for Applied Macroeconomic Research is part of Sao Paulo School of Economics;
- The main goal of the Center is to boost Applied Macroeconomic Research at the School;
- Our web-page is www.cemap.fgv.br/;

Our projects

- Observatory on Exchange Rate;
- Forecast Lab;
- Spatial Econometrics;
- Time Series Econometrics;

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Macroeconomics and Regional Data:

- Macroeconomic shocks affect sectors and regions with different intensities and delays;
- Population and economic activities tend to be clustered in specific areas due economies of scale and agglomeration.
- The population tends to be located near sea coast compared with interior continental areas.
- Sector activities tend to be concentrated in some areas or cities.

Motivation

- Macroeconomic models tend to disregard regional and spatial economic activity issues due to
 - either their aggregate nature bias
 - or complexity of theme.
- However for a specific region, it is important to understand how economic activity variance can be decomposed into
 - common factors (possibly macroeconomic factor) and
 - idiosyncratic factors;

Goals

- We want to model possible interaction between macroeconomic shocks and regional employment dynamics;
- Assess possible different responses of regional employment to the same shock;
- Generate scenarios for regional employment evolution in Brazil bases on a well specific time series model;

Possible innovation

- Apply GVAR to link macroeconomic with regional dimension in Brazil;
 - Deal with curse of dimensionality;
- Use of IBGE Survey on economic links of Brazilian economy to construct weights in GVAR;

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Global VAR approach

- Global autoregressive vector (GVAR) methodology was made by Pesaran, Weiner and Schuermann (2004), and improved on by Dees, Mauro, Pesaran and Smith (2007);
- Chudik and Pesaran (2011) - Analyzed the curse of dimensionality in the case of large dynamic linear systems.
- Infinite-dimensional VAR could be well approximated by a set of finite dimensional small-scale models that can be consistently estimated separately.
- Spatial analysis application is a natural option of this framework;

The Global VAR model as a two-step procedure

- First stage: Specific models of each region are estimated conditioned on external influences from other regions.

$$x_{it} = a_{i0} + a_{i1}t + \sum_{j=1}^{p_i} \Phi_{ij} x_{i,t-j} + \Lambda_{i0} x_{it}^* + \sum_{l=1}^{q_i} \Lambda_{il} x_{it-l}^* + u_{it} \quad (1)$$

$$x_{it}^* = \sum_{j=0}^N W_{ij} x_{jt}$$

- And they can be restated as:

$$A_{i0} W_i x_t = a_{i0} + a_{i1}t + \sum_{l=1}^p A_{il} W_i x_{t-l} + \varepsilon_t \quad (2)$$

The Global VAR model as a two-step procedure

- Second Stage: The VAR models of each region are grouped into a single Global VAR model.

$$G_0 = \begin{bmatrix} A_{00}W_0 \\ A_{10}W_1 \\ \vdots \\ A_{N0}W_N \end{bmatrix}; \quad G_I = \begin{bmatrix} A_{0I}W_0 \\ A_{1I}W_1 \\ \vdots \\ A_{NI}W_N \end{bmatrix}; \quad a_0 = \begin{bmatrix} a_{00} \\ a_{10} \\ \vdots \\ a_{N0} \end{bmatrix}; \quad a_1 = \begin{bmatrix} a_{01} \\ a_{11} \\ \vdots \\ a_{N1} \end{bmatrix}; \quad \varepsilon_1 = \begin{bmatrix} \varepsilon_{0t} \\ \varepsilon_{1t} \\ \vdots \\ \varepsilon_{Nt} \end{bmatrix};$$

$$G_0 x_t = a_0 + a_1 t + \sum_{l=1}^p G_l x_{t-l} + \varepsilon_t$$

$$F_i = G_0^{-1} G_i; \quad b_0 = G_0^{-1} a_0; \quad b_1 = G_0^{-1} a_1; \quad e_t = G_0^{-1} \varepsilon_t;$$

$$x_t = b_0 + b_1 t + \sum F_i x_{t-i} + e_t$$

Selected papers on the subject

- Oliveira and Carneiro (2001) - Analysis of the dynamics of the employment level of several Brazilian states in relation to national employment with the objective of verifying a long term relationship.
- Camila Kraide Kretzmann and Marina Silva da Cunha (2009) - Analysis of the formal labor market between metropolitan and non-metropolitan areas in Brazil.
- Norbert Schanne (2015) - Analysis of the regional labor market in Germany, focusing on forecasting regional labor markets. She used the GVAR methodology to perform a prospective analysis of the German economy.
- All these papers confirms the existence of links among regions;

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Brazilian Economy Structure



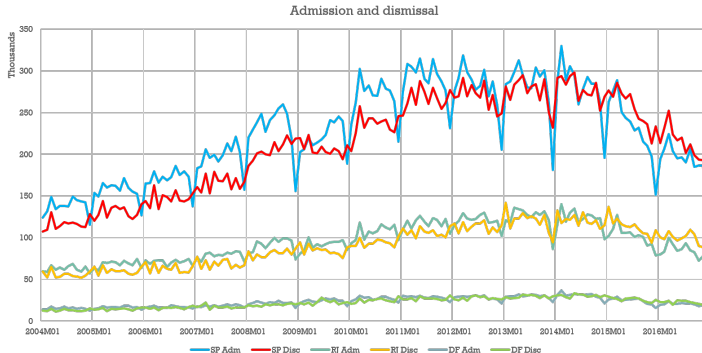
- South and South East - Industrial, services and most developed part of the country
- North East - Agricultural and tourism
- North - Agricultural - Amazon forest;
- Central - New agricultural frontier;

Figure: Brazilian Regions

Data

Variable	Source	Frequency	Range
Brazilian Industrial Production	IBGE	Monthly	2004-01 to 2016-12
Admissions and Dismissals	CAGED	Monthly	2004-01 to 2016-12
Meso-regions and municipalities	IBGE	N/A	as of 2015
Connections between regions	IBGE	N/A	as of 2007

Some Graphics



Model

- The units in the model are the mesoregions defined by the Brazilian Institute of Geography and Statistics.
- The labor market variables are the numbers workers admitted and dismissed from the labor market.

$$x_{it} = \begin{bmatrix} Adm_{it} \\ Dis_{it} \end{bmatrix}; \quad x_{it}^* = \begin{bmatrix} Adm_{it}^* \\ Dis_{it}^* \end{bmatrix}$$

$$Adm_{it}^* = \sum_{j=1}^N W_{ij} Adm_{jt} \quad Dis_{it}^* = \sum_{j=1}^N W_{ij} Dis_{jt}$$

Simple Macroeconomic Model

- Macroeconomic model contains Brazilian production industrial index and employment admissions and dismissals with feedback from regions:

$$\omega_t = [\ln(IndProd)]$$

$$\Delta \omega_t = \Phi_{\omega 0} + \sum_{l=1}^{k_1} \Phi_{\omega l} \Delta \omega_{t-l} + \sum_{l=1}^{k_2} \Lambda_{\omega l} \Delta x_{i,t-l}^* + \eta_{\omega t}$$

Our weight matrix choice

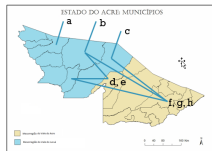


Figure: Response to changes in growth of industrial production

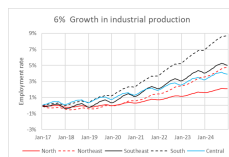
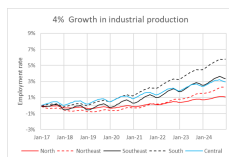
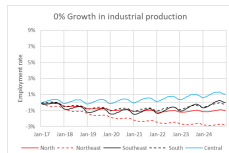
- Survey of Brazilian Bureau of Geographic and Statistics on economic links of regions;
- Links available at all municipalities;

$$W_{blue} = \frac{\begin{bmatrix} c_{i1} & \dots & c_{iN} \end{bmatrix}}{\sum_{j=1}^N c_{ij}} \quad W_{blue} = \frac{\begin{bmatrix} 5 & 3 \end{bmatrix}}{8} = \begin{bmatrix} 0.625 & 0.375 \end{bmatrix}$$

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Brazilian Five Regions Response



- South (0.18)
- North East (0.16)
- South East (0.1)
- North (0.06)
- Central (0.06);

Table: Response to changes in growth of industrial production

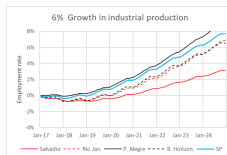
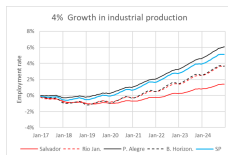
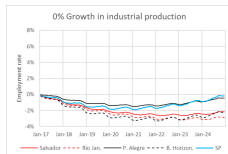
Southeast States Response



- Rio de Janeiro (0.19)
- Espírito Santo (0.09)
- São Paulo (0.08)
- Minas Gerais (0.07);

Table: Response to changes in growth of industrial production

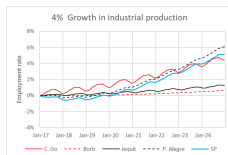
Selected meso regions



- Rio de Janeiro (0.19)
- Porto Alegre (0.19)
- Belo Horizonte (0.17)
- São Paulo (0.16)
- C. Goiás (0.07)
- Salvador (0.11)

Table: Response to changes in growth of industrial production

Selected Mesoregions



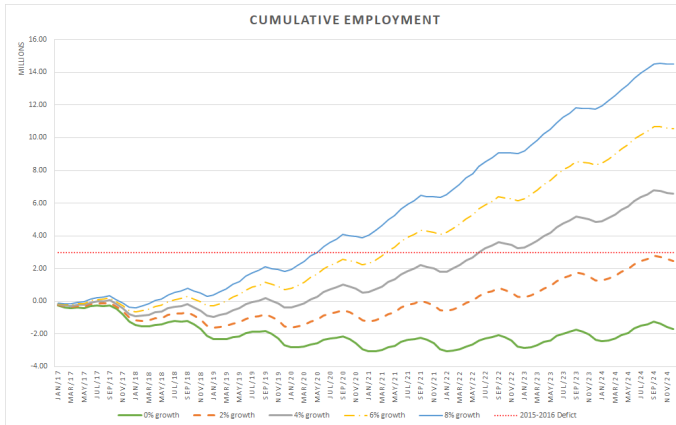
- Porto Alegre (0.19)
- São Paulo (0.16)
- C. Goiás (0.07)
- Borborema (0.02)
- Jequit. (0.02);

Table: Response to changes in growth of industrial production

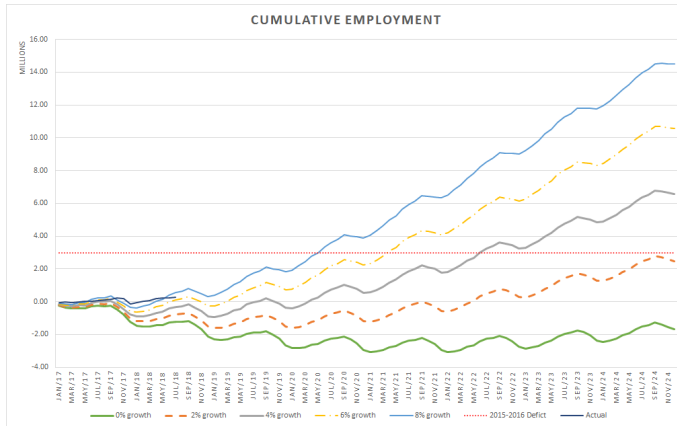
Scenario for growth

- Several scenarios were estimated to assess the impact of industrial production on the labor market.
 - Annual growth of 0%: A negative employment balance for at least the next 8 years.
 - Annual growth of 2%: A positive accumulated balance of 2.5 million job positions for 8 years ahead.
 - Higher growth scenarios (4%, 6% and 8%) show significant improvements in the labor market.

Conditional predictions from different scenarios



Forecasts vs. Outcome



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Limitations

- Macroeconomic model can be improved to incorporate more variables and disentangle possible different types of shocks;
- Different weights strategies can be tried and compared;
- Evaluation of forecast performance.

Extensions

- Work at higher disaggregated level but “municipality split epidemic” in Brazil must be addressed - Comparable minimum areas;
- Models can be improved for each unity using *Autometrics* algorithm similar to Ericsson et al. (2012);
- Using MINIMAL algorithm to unveil cointegration relationship among cities to construct a “expanded” GVAR;

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Small Macro Model

- Starting Point:
 - VARX with twelve lags for all variables plus error correction terms;
 - Centered Seasonal Dummies;

X_1 - Net Employment;

X_2 - Dismissals;

X_3 - Δ In Industrial Production;

X_4 - Δ In Selic

Instability Points

DI: First Difference Impulse Dummy

I: Impulse Dummy

SI: Step Dummy

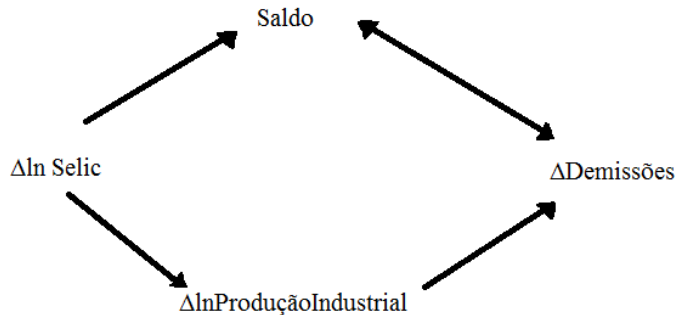
- DI:2001(12)
- I:2003(3)
- I:2008(12)
- I:2018(6)
- S1:2012(2)
- S1:2012(3)
- S1:2012(12)
- S1:2013(1)
- S1:2013(11)
- S1:2013(12)
- S1:2014(2)

Temporal Sequence

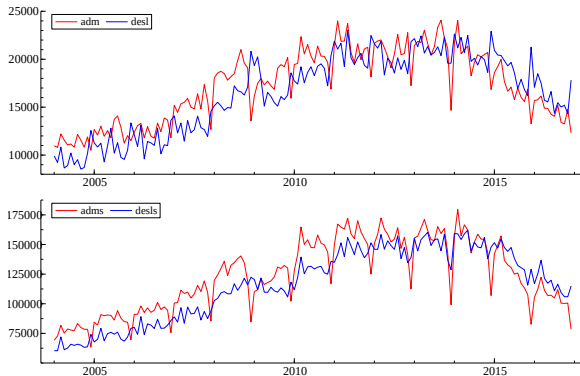
Past	->	Present	Statistics	Distribution	p-value
X1	->	X1	10.97	$\chi^2(1)$	[0.0009]
X2	->	X1	31.43	$\chi^2(3)$	[0.0000]
X3	->	X1	32.03	$\chi^2(7)$	[0.0000]
X4	->	X1	19.81	$\chi^2(4)$	[0.0005]
X1	->	X2	28.51	$\chi^2(2)$	[0.0000]
X2	->	X2	88.82	$\chi^2(3)$	[0.0000]
X3	->	X2	19.61	$\chi^2(7)$	[0.0065]
X1	->	X3	10.82	$\chi^2(2)$	[0.0045]
X3	->	X3	28.48	$\chi^2(7)$	[0.0002]
X4	->	X3	37	$\chi^2(4)$	[0.0000]
X4	->	X4	64.63	$\chi^2(4)$	[0.0000]

X1 - Saldo; X2 - Demissões-Var; X3 - Δ In Industrial Production; X4 - Δ In-Selic

Granger Causality

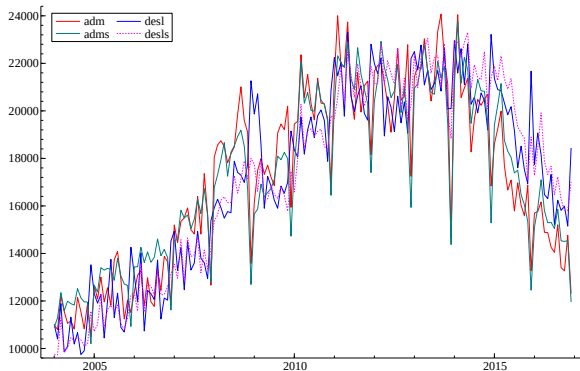


Vale do Paraíba



- Admissions and dismissal cointegrate.

Do they cointegrate?



- But more than two cointegrated vectors!

Conditional model for Admissions: Vale do Paraíba

- Starting Point:
 - VARX with twelve lags for all variables plus error correction terms;
 - Centered Seasonal Dummies;
- Possible overfitting;

Final model conditional model for Dismissals: Vale do Paraíba

$$\begin{aligned}
 \text{Dadm} = & - 0.1615 \text{ Dadm}_{t-1} - 0.1213 \text{ Dadm}_{t-2} - 0.1228 \text{ saldo}_{t-1} \\
 & \quad (0.0423) \quad (0.0369) \quad (0.0607) \\
 & - 0.03239 \text{ saldo-adms}_{t-1} + 0.03253 \text{ saldo-desls}_{t-1} + 0.1146 \text{ Dadms}_t \\
 & \quad (0.00998) \quad (0.0103) \quad (0.00568) \\
 & + 0.001022 \text{ Ddesls}_t + 0.01186 \text{ Dadms}_{t-7} - 0.02118 \text{ Ddesls}_{t-3} \\
 & \quad (0.00875) \quad (0.00437) \quad (0.00811) \\
 & + 1039 \text{ CSeasonal}_{t-10} + 2268 \text{ DI:2012(11)}_t + 1710 \text{ DI:2013(9)}_t \\
 & \quad (238) \quad (550) \quad (518) \\
 & - 94.59 \\
 & \quad (304)
 \end{aligned}$$

- “Global Factors” explain part of the dynamics of admissions;

Conditional model for Admissions: Vale do Paraíba

$$\begin{aligned}
 Ddesl = & - 0.2092 \text{ Dadm}_{t-1} - 0.1881 \text{ Dadm}_{t-2} + 0.4811 \text{ saldo}_{t-1} \\
 & (0.0462) \quad (0.0403) \quad (0.0663) \\
 & + 0.06565 \text{ saldo-adms}_{t-1} - 0.06865 \text{ saldo-desls}_{t-1} - 0.03675 \text{ Dadms}_t \\
 & (0.0109) \quad (0.0113) \quad (0.00621) \\
 & + 0.1568 \text{ Ddesls}_t + 0.01593 \text{ Dadms}_{t-7} - 0.02962 \text{ Ddesls}_{t-3} \\
 & (0.00955) \quad (0.00477) \quad (0.00886) \\
 & - 209.2 \text{ CSeasonal}_{t-10} + 433.5 \text{ DI:2012(11)}_t + 409.8 \text{ DI:2013(9)}_t \\
 & (260) \quad (601) \quad (566) \\
 & - 216.3 \\
 & (332)
 \end{aligned}$$

- “Global Factors” also explain part of the dynamics of dismissals;

Good results on specification tests

- Vector AR 1-7 test: $F(28,230) = 1.6157 [0.0306]^*$
- Vector Normality test: $\chi^2(4) = 9.7889 [0.0441]^*$
- Vector Hetero test: $F(69,350) = 1.3114 [0.0619]$
- Vector Hetero-X test: $F(204,216) = 1.2511 [0.0524]$
- Vector RESET23 test: $F(8,250) = 1.0066 [0.4315]$

Putting all the pieces together

- Solving the model using GVAR under this approach;
- We can simulate how changes on variables such interest rate will affect employment at regional level;
- And forecast them as well;

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Some tentative conclusions

- South, Southeast and Northeast have shown a stronger response to shocks in Brazilian aggregate industrial production;
- Macro factor does not play strong influence on North and Central meso regions;
- Formal employment will tend to recover slowly in Brazil in near future unless high growth rates start to be observed;